Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

Data Science Center, UCLA Library

UCLA OSPO Lead, UC OSPO Network

The invisible layer

Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **67%** say their research would be impossible without it.

The people keeping it running are increasingly called **Research Software Engineers (RSEs)**, a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **$8.8 trillion**.

Hettrick et al. (SSI, 2014/2022) · Hoffmann, Nagle & Zhou (2024) doi:10.2139/ssrn.4693148

Jupyter

Python

R

Git

Spark

Pandas

NumPy

Docker

Ray

ggplot2

dplyr

Snakemake

Nextflow

The UC research-to-infrastructure pipeline

Data professionals built open data infrastructure: FAIR, DMPs, curation, provenance.

FAIR4RS is the same inflection point for software — and UC has been showing the way for 50 years.

BSD Unix · Ceph · Apache Spark · Jupyter · RISC-V · Ray

The pipeline from university research to global infrastructure is not rare or accidental.

Barker et al. (2022) doi:10.1038/s41597-022-01710-x

UC Berkeley

Jupyter

Apache Spark

RISC-V

Ray

BSD Unix

UC Santa Cruz

Ceph

UCSC Genome Browser

UC San Diego

Cytoscape EEGLAB

UCSD Pascal

UCLA

Processing

Named Data Networking

tz database

UC San Francisco

ChimeraX

OpenMM

UC Davis

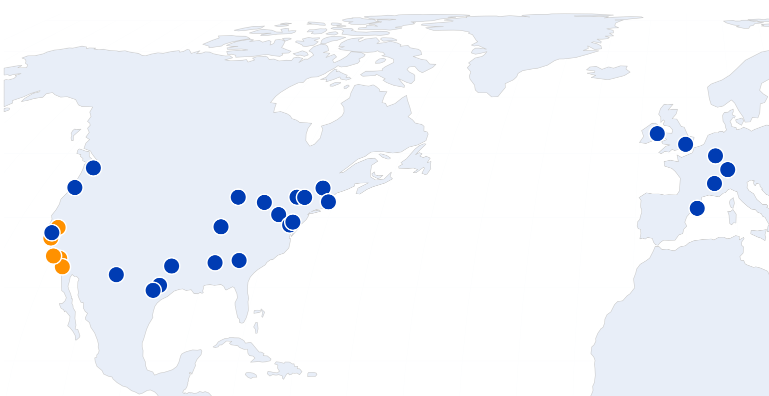
sourmash

ERPLAB

What is an academic OSPO?

An **Open Source Program Office** supports, governs, and promotes open source within an institution.

- Originated in industry: ~77% of large tech companies have one
- First academic OSPO: Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



🟠 UC OSPO Network · 🔵 CURIOSS member

World map showing 34 CURIOSS member institutions. Six U

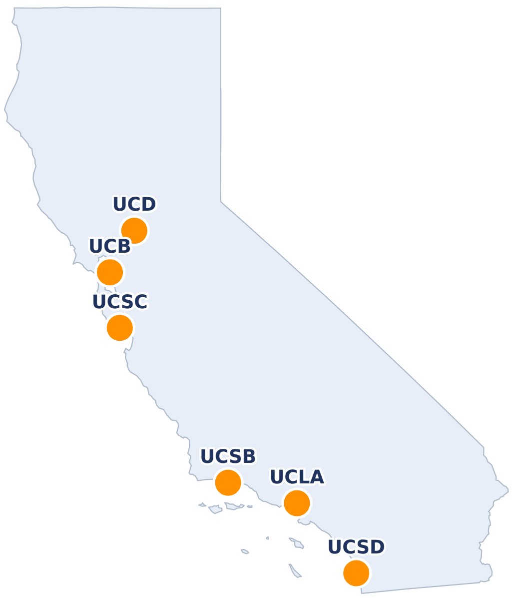
The UC OSPO Network

A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
- Launched April 2024, funded by the Alfred P. Sloan Foundation
- **$1.85M** in external funding to date
- Lead: UC Santa Cruz, the first OSPO in a large state university system
- Serves 280,000+ students and 25,000+ faculty

Acts as a **neutral convener**: no single campus, company, or grant

cycle can pull the work away.
Ruff (2026) UC Open Summit · youtu.be/eBriL3CDNeo



Map of California showing six UC OSPO Network campuses:

Discovery · Sustainability · Education

🔭

Discovery
Mapping who does what across the UC system

- 200,000+ repos scanned; ~52,000 institutionally affiliated
- 294-respondent system-wide survey
- UC Open Repository Browser (UC ORB)

🌱

Sustainability
Keeping projects and communities healthy

- **58%** of UC contributors are also maintainers
- Cross-campus licensing working group (UCOP + tech transfer)
- Project health assessment (OSSPREY)

📖

Education
Coordinated training across

campuses

- Gap analysis → curriculum inventory → learning pathways
- Published at **ucospo.net/education**
- Coordinated with Carpentries infrastructure

Gomez et al. (2025) arxiv:2506.18359 · Scarlett et al. (2025) doi:10.31235/osf.io/p8bx6_v1

Lessons from two years

What the network model enables that single campuses cannot:

1. **Shared staffing**: community manager, licensing specialist, technical roles
2. **Policy leverage**: the network has standing with UCOP; individual campuses do not
3. **Data at scale**: the GitHub and survey analyses require system-wide scope to mean anything
4. **Equity**: smaller campuses get services they could not build alone
5. **A replicable model**: $1.85M produced something other state systems can follow

Each campus contributes unique expertise; the network routes it to where it's needed.

Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

You’re already doing this work

If you do technical data services, researchers are already bringing you these questions:

- “What license should I use for this code?”
- “Is this library still maintained? Should I depend on it?”
- “How do I make this reproducible and citable?”
- “My funder wants a software management plan.”
- “How do I accept contributions from collaborators?”

That is OSPO work. The frame makes it legible to your institution.

The network has already built the resources (Laura Langdon, UC OSPO):

`README` · `LICENSE` · `CONTRIBUTING` ·

`CODE_OF_CONDUCT` · `CITATION.cff` + Zenodo DOI · **Growing a Community** · **Carpentries Incubator lesson**

Raise the floor, not just solve the ticket
Fixing their immediate problem gets them through the week. Moving them from “it runs on my machine” to maintainable, licensed, citable software gets their whole lab there, and their students after them.
You don’t need a formal OSPO to do this. You need the frame.

OSPO heuristics already in your toolkit: health signals · bus factor · governance files · FAIR4RS · software DMPs · contributor conventions · community governance

Learn more / connect

UC OSPO Network: ucospo.net

- Education site: **ucospo.net/education**
- Mailing list, Slack, events calendar

Global network: curioss.org, 34 academic OSPOs worldwide and growing

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> 🖥️ Slides: **tinyurl.com/iassist2026**
> 📄 Handout: **tinyurl.com/iassist2026-handout**

QR code linking to tinyurl.com/iassist2026 — slides for this presentation

tinyurl.com/iassist2026

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Speaker notes